

Bluestock MF

Mutual Fund Analytics — India

An end-to-end analytics project covering data ingestion, ETL, validation, exploratory analysis, performance metrics, dashboard reporting, and investor-oriented recommendations.

PROJECT FOCUS	Indian equity mutual fund comparison
DATA WORKFLOW	Generate / ingest → clean → load → analyze → report
TECHNOLOGY	Python, pandas, NumPy, Matplotlib, Seaborn, Plotly, SQLAlchemy, Jupyter
LIVE DATA	mfapi.in for selected scheme NAVs
PRIMARY USERS	Investors, fund analysts, and research stakeholders

Prepared from the supplied project README, capstone report, and presentation materials.

1. Executive Summary

Bluestock MF is designed as a repeatable mutual fund analytics workflow for the Indian equity market. The project turns structured fund data and selected live NAV feeds into comparable metrics, visual analysis, and investor-oriented reporting. The supplied materials position the solution for scheme screening, SIP suitability analysis, performance monitoring, and portfolio selection.

The analysis combines NAV history, returns, expense ratios, AUM, benchmark information, holdings, SIP records, fund-manager data, and risk metrics. The ETL workflow emphasizes validation and repeatability so that the same process can be rerun when new data becomes available.

Key reported statistics are an average 1-year return of 13.67% and an average expense ratio of 1.30%. The report also highlights positive-alpha schemes, moderate expense ratios, and stable AUM behavior as useful signals when screening funds. These are analytical indicators, not personalized investment advice.

Project Scope Note

The supplied README documents a 20-scheme master universe, while the supplied report appendix states 40 schemes and 139,760 NAV observations. This final version preserves both source statements rather than silently reconciling them: the dataset inventory is described as documented in the README, while aggregate report statistics are quoted from the report appendix.

2. Background and Objectives

Indian mutual fund selection can become difficult when investors compare schemes using only a single return number. The project therefore frames fund selection as a multi-factor analytics problem involving return, risk, cost, benchmark-relative performance, and AUM behavior.

Objectives

- Build a repeatable pipeline for mutual fund data ingestion and analysis.
- Validate scheme identifiers and maintain consistent datasets across analysis stages.
- Compare schemes using returns, alpha, expense ratios, AUM, NAV behavior, and risk metrics.
- Create clear visual outputs for stakeholder and investor review.
- Provide practical screening recommendations while documenting limitations and future improvements.

Business Uses

The supplied report identifies four core uses: scheme screening, SIP suitability analysis, performance monitoring, and portfolio selection. For fund research teams, the same workflow can support summary metrics, rankings, and visual comparisons without rebuilding the analysis manually each time.

3. Data Sources and Project Architecture

The README describes a structured project containing raw data, processed-data space, notebooks, SQL analysis, dashboard outputs, reports, and Python pipeline modules. The main ingestion and live-NAV scripts are separated from dataset generation, which keeps the workflow modular.

Dataset Inventory

Dataset	Rows	Purpose
fund_master.csv	20	Scheme metadata, fund house, category, risk grade, benchmark
nav_history.csv	10,000	Daily NAV history for 20 schemes, 2015–2024
portfolio_holdings.csv	200	Top 10 stock holdings per scheme
sip_data.csv	600	Monthly SIP transactions, Jan 2020–Dec 2024
returns_data.csv	20	1m / 3m / 6m / 1y / 3y / 5y / inception returns
benchmark_data.csv	500	Nifty 50, Nifty 100, BSE 500, Nifty Midcap
aum_data.csv	720	Monthly AUM and folio count, 2022–2024
expense_ratio.csv	20	Direct vs regular plan TER
fund_manager.csv	20	Manager name, experience, qualification
risk_metrics.csv	20	Std dev, beta, Sharpe, alpha, R ² , Sortino

Live NAV files documented in the README were fetched from mfapi.in for six schemes: HDFC Top 100 Direct (125497), SBI Bluechip (119551), ICICI Bluechip (120503), Nippon Large Cap (118632), Axis Bluechip (119092), and Kotak Bluechip (120841).

4. ETL and Data Quality Workflow

The pipeline follows a practical sequence: generate or ingest datasets, validate identifiers, clean and normalize records, load structured tables, compute analytics, and generate deliverables. The README also documents an orchestrator, `run_pipeline.py`, which can list or execute individual stages.

Stage	What happens	Output
1. Ingestion	Load CSV datasets and explore fund master data.	Structured source tables
2. Validation	Check AMFI scheme-code coverage and key fields.	Validated identifiers
3. Cleaning	Parse dates, handle NAs, review outliers, normalize categories.	Consistent analytics data
4. Load	Load cleaned tables for structured analysis.	Analysis-ready tables
5. Analytics	Compute returns, alpha, risk, expense, AUM and comparison metrics.	Metrics + rankings
6. EDA	Create NAV, return, expense, and AUM visuals.	Charts / dashboard inputs
7. Deliverables	Build the PDF report and presentation.	Final stakeholder outputs

Data-quality note: the README states that AMFI code coverage is 100% across the 20 master codes and corresponding NAV history. It also warns that live API scheme names can differ from labels because AMFI codes may represent direct/regular, growth/IDCW, or other scheme variants; the API response's `scheme_name` should therefore be verified.

5. Exploratory Data Analysis

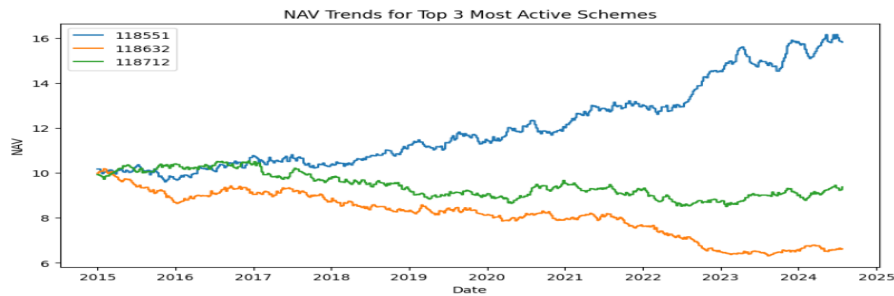
The supplied EDA identifies differences in NAV histories, expense ratios, and return profiles. The report specifically notes that expense-ratio dispersion and NAV volatility help explain scheme-level performance variation.

NAV Trends

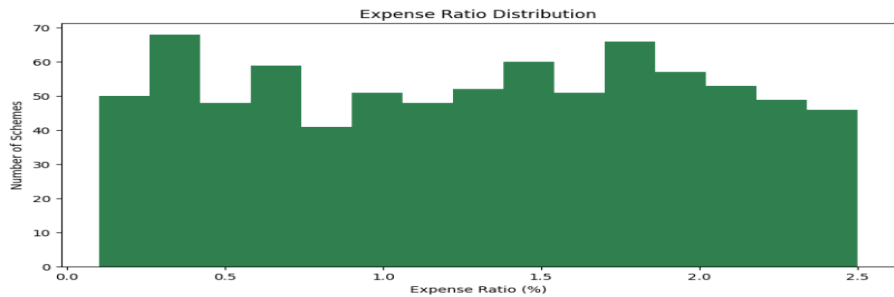
Exploratory Data Analysis

Exploratory analysis reveals that sample funds have varied NAV histories, expense ratios, and return profiles. Expense ratio dispersion and NAV volatility explain much of the scheme-level performance variation. Top fund rankings by alpha and AUM highlight different risk-return tradeoffs for investors.

Nav Trends



Expense Ratio Distribution



The NAV chart on page 8 of the supplied report shows three example scheme series diverging substantially over 2015–2024, illustrating why historical trajectory and volatility matter alongside point-in-time returns.

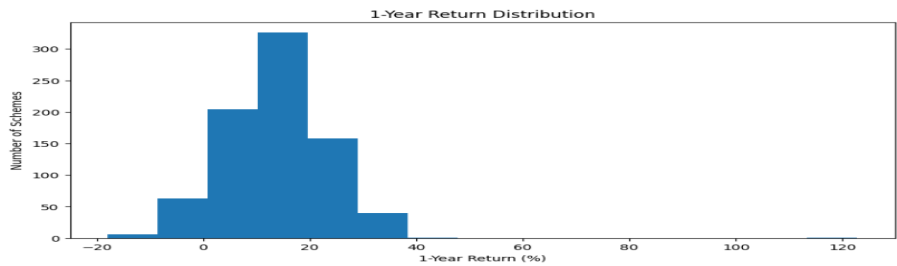
6. Performance Findings

The performance section evaluates 1-year return distribution, benchmark-relative alpha, expense-ratio impact, and AUM momentum. The stated objective is to identify schemes showing attractive risk-adjusted outcomes and consistent growth while keeping downside risk under consideration.

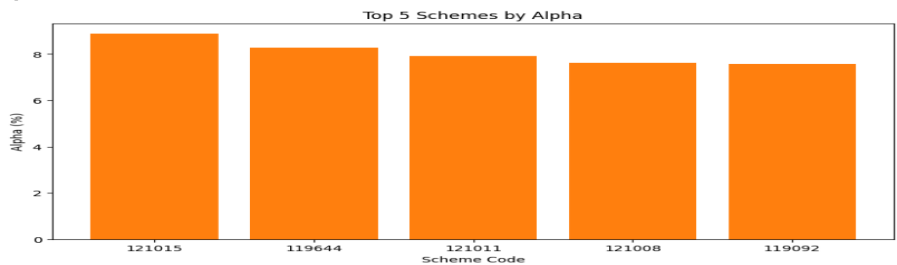
Performance Findings

Performance metrics include 1-year returns, alpha, expense ratio impact, and AUM momentum. This analysis identifies schemes with positive benchmark-relative performance and attractive risk-adjusted outcomes. The findings support investor selection of schemes with consistent growth and controlled downside risk.

Return Distribution



Top Alpha



The return-distribution chart and top-alpha chart are reproduced from the supplied report. The top-alpha ranking lists scheme codes 121015, 119644, 121011, 121008, and 119092, with the leading value visually just under 9%.

7. Cost, AUM, and Dashboard Insights

Expense ratio is treated as a practical cost driver: lower cost can improve investor outcomes when other characteristics are comparable, but the project does not claim that expense ratio alone determines fund quality. The report's aggregate average expense ratio is 1.30%.

AUM Momentum

Recommendations

Prioritize funds with positive alpha, moderate expense ratios, and stable AUM growth. Schedule regular analytics updates and extend the dashboard with interactive filters and scenario simulations. Future work should include Monte Carlo modeling, optimization, and automated investor reporting.

The AUM chart on page 12 shows substantial month-to-month movement for scheme 121018 across 2022–2024. The supplied recommendations therefore emphasize monitoring AUM stability rather than treating a single AUM value as sufficient evidence.

Dashboard Design

The dashboard combines NAV trends, expense-ratio distribution, performance rankings, and AUM analytics using the same analytics dataset. This is intended to support rapid scheme screening and stakeholder review.

8. Recommendations

Based strictly on the supplied project conclusions, the screening approach should prioritize funds that combine positive alpha, moderate expense ratios, and stable AUM growth. These indicators should be evaluated together with downside risk rather than used independently.

Recommendation	Practical interpretation
Prioritize positive alpha	Use benchmark-relative performance as one signal of value added by the scheme.
Control cost	Prefer moderate expense ratios when comparing otherwise similar funds.
Monitor AUM behavior	Look for sustained or stable AUM patterns instead of relying on a single observation.
Review downside risk	Combine return and alpha with volatility and other risk metrics before making a selection.
Automate refreshes	Schedule regular data and analytics updates so the dashboard remains current.

Future Work

The supplied report recommends live daily AMFI feed integration, a wider scheme universe, direct-vs-regular plan differentiation, interactive dashboard filters, scenario simulation, Monte Carlo modeling, portfolio optimization, and automated investor reporting.

9. Limitations and Data Governance

The current analysis is not a production investment engine. The supplied report explicitly notes that it uses sample CSV data and a limited subset of schemes. It recommends live daily AMFI integration and a wider scheme universe for production use.

The report also notes that direct-versus-regular plan differentiation is not included in the current model. The README separately warns that API scheme labels can differ from local labels, so scheme-code and scheme-name validation should remain part of the pipeline.

Conclusion

Bluestock MF demonstrates how a structured ETL and analytics pipeline can turn mutual fund data into a practical decision-support workflow. Its strongest contribution is the repeatability of the process: the same stages can be rerun, validated, analyzed, and converted into stakeholder deliverables. The results support multi-factor fund screening rather than one-dimensional ranking.

The project is therefore best viewed as an analytics foundation. With live AMFI integration, broader coverage, plan-level differentiation, interactive scenario analysis, and stronger portfolio optimization, it can evolve into a more comprehensive investor-research platform.

Key Report Statistics

Metric	Reported value
Average 1-year return	13.67%
Average expense ratio	1.30%
Schemes in report appendix	40
NAV observations in report appendix	139,760

Source basis: supplied README.md, Final_Report.pdf, and Bluestock_MF_Presentation.pptx.